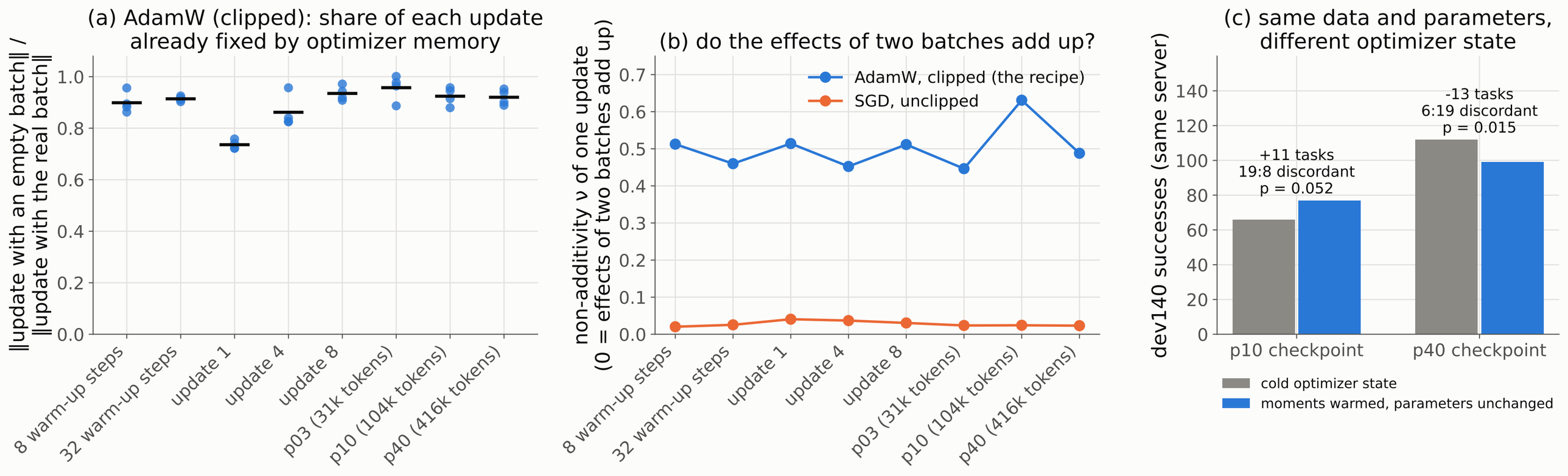
(a) AdamW (clipped): share of each update already fixed by optimizer memory
‖update with an empty batch‖ / ‖update with the real batch‖
1.0
0.8
0.6
0.4
0.2
0.0
8 warm-up steps
32 warm-up steps
update 1
update 4
update 8
p03 (31k tokens)
p10 (104k tokens)
p40 (416k tokens)
(b) do the effects of two batches add up?
non-additivity ν of one update (0 = effects of two batches add up)
AdamW, clipped (the recipe)
SGD, unclipped
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
8 warm-up steps
32 warm-up steps
update 1
update 4
update 8
p03 (31k tokens)
p10 (104k tokens)
p40 (416k tokens)
(c) same data and parameters, different optimizer state
dev140 successes (same server)
+11 tasks
19:8 discordant
p = 0.052
-13 tasks
6:19 discordant
p = 0.015
140
120
100
80
60
40
20
0
p10 checkpoint
p40 checkpoint
cold optimizer state
moments warmed, parameters unchanged